ELECTRICAL MACHINES · CH3

Regulation, efficiency, 3-phase transformers

Cheat sheet · Day 05 · Wed 28 Oct · pack M5

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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TRANSFORMER LOAD / REGULATION / EFFICIENCY (per phase, simplified ckt ref. primary)
 R = R1+R2'   X = X1+X2'   shunt Rc || jXm across V1 (input)
 Star: V1 = VL/sqrt3, I1 = IL     Delta: V1 = VL, I1 = IL/sqrt3   (assume star)
 V2' = VL,rated/sqrt3 /_0         S = P/pf  or  S = sqrt(P^2+Q^2)
 I2' = S/(3 V2') /_ -acos(pf) lag, + lead   (Q<0 => leading)
 V1 = V2' + I2'(R + jX)     PHASOR sum -> rectangular -> polar; line = sqrt3|V1|
 Regulation VR = 100(V2NL - V2FL)/V2FL = 100(|V1| - |V2'|)/|V2'|   (V2NL' = V1)
   check: |V1|-|V2'| ~ I2'(R cos(phi) +/- X sin(phi))  (+ lag, - lead; lead can give
 PCU = 3 I2'^2 R  (use I2', not I1)  ~ x^2 PCU,FL   (x = per-unit load)
 PFE = 3 V1^2 / Rc  (use V1, not V2')  ~ constant, ~ V^2
 eta = Pout/(Pout+PCU+PFE)   or (Pin - losses)/Pin if Pin given
 Max eta: PCU = PFE -> x = sqrt(PFE/PCU,FL); eta_max = x S pf/(x S pf + 2 PFE)
OPEN CIRCUIT (I2' = 0): Z = Rc || jXm; I1 = V1/|Z|; P = 3V1^2/Rc; Q = 3V1^2/Xm
 V halved: I x0.5, P x0.25, Q x0.25, pf x1 (model). Real core: I falls MORE (less sat
SHORT ANSWERS
 Cu loss: I^2R in windings, series R, ~ (output)^2.  Fe loss: hysteresis + eddy, Rc,
 No-load I != 0: magnetising mmf (finite mu, Xm, lags 90deg) + core-loss current (Rc)
 Saturation: B ~ V/(N f); near knee mu falls -> Xm falls -> I rises fast, distorted.
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